Nidhir Bhavsar

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PROFILE

AI Engineer with 5+ years of experience building reliable AI systems across research and industry environments. I design structured document and data processing pipelines, multi-agent workflows, and evaluation frameworks that make complex, contract-heavy processes measurable and scalable. My focus is robustness, schema validation, cost-aware deployment, and production reliability across Azure and AWS. I combine research-driven experimentation with practical system design to deliver AI solutions that are both interpretable and operationally stable.

WORK EXPERIENCE

1Alpha GmBH, AI Engineer — Dec 2025 – Present | Munich, Germany

- Drove early architectural improvements to agentic LLM workflow by introducing LangGraph-based orchestration, improving robustness, fallback handling, and runtime performance for Azure-hosted LLMs; owned production readiness via CI/CD automation, monitoring, fallback strategies and deployment reliability for Azure-hosted LLM systems.
- Improved LLM-driven document processing pipelines for complex PDF-based (insurance contracts) workflows by refactoring schema-based extraction logic, implementing structured validation layers, and stabilizing distributed task orchestration (Celery/Redis); enhanced observability (Langfuse) to ensure traceable and reliable AWS deployments.

Patrick Bunk Wiescke, AI Engineer — Oct 2024 – Nov 2025 | Berlin, Germany

- Evaluated multi-step LLM agents in computer-use environments; architected a LangGraph-based multi-agent system enabling web interaction, automated email triage via MCP, and intent-driven ServiceNow ticket creation, achieving ~30% performance gains through meta-learning and SOP-guided behaviors.
- Designed and optimized a structured data extraction and validation pipeline for complex client PDF forms, leveraging iterative prompt engineering, schema-based validation, and LLM-as-a-judge evaluation loops to achieve near-100% accuracy across all document types.
- Built LLM observability and evaluation infrastructure using LangSmith, enabling end-to-end tracing, telemetry, human-in-the-loop interrupts, and continuous workflow evaluation, significantly improving system debuggability and reliability.
- Implemented multi-provider LLM orchestration with intelligent routing, retries, and fallbacks across OpenAI, Google, Anthropic, and Azure-deployed OpenAI models; optimized Azure-hosted models to close performance gaps with experimental baselines

Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI-SLT), Student Research Assistant — Jan 2023 – Sep 2024 | Berlin, Germany

- Developed and evaluated a low-resource information extraction pipeline using BERT-based models and few-shot learning (prototypical networks, episodic clustering). Achieved performance comparable to fine-tuned domain-specific models & LLMs across multiple domains.
- Maintained and deployed open-source LLMs in an on-premises sandbox with a ChatGPT-like UI (FastAPI, React), collaborating with members of other DFKI teams. Served HuggingFace models via vLLM and enabled streaming, logging, and controlled evaluation on a GPU cluster (SLURM).
- Led an industry supply-chain optimization project focused on identifying material alternatives in the Bill of Materials (BoM) by developing a RAG based application. Integrated open-source LLM (LLaMA2) with a vector database and vLLM for efficient inference; validated the pipeline for high accuracy and minimal error using RAG evaluation (RAGAs), and deployed the final application on AWS via a CI/CD pipeline using GitHub Actions.

Foundations of Computational Linguistics Lab, University of Potsdam, Student Researcher — Oct 2023 – Jun 2024 | Potsdam, Germany

- Led evaluation of LLM performance trade-offs across model scale, instruction-tuning data quality, efficiency techniques (e.g., quantization), and access methods for goal-oriented tasks, using automated benchmarking workflows.
- Designed and executed scalable LLM evaluation pipelines across proprietary & open-source models (e.g., GPT-4, Claude Sonnet, LLaMa-3), enabling data-driven model selection on performance, cost, latency, and deployment constraints; showed that data quality could rival model scale.

EDUCATION

University of Potsdam, M.Sc. Cognitive Systems GPA: 1.4/1 — Oct 2022 – Sep 2025 | Potsdam, Germany

Navrachana University, B.Tech. Computer Science & Engineering GPA 8.2/10 — Jun 2018 – Jun 2022 | Vadodara, India

SKILLS

Languages: Python, JavaScript, TypeScript

Framework & Tools: LangGraph, LlamaIndex, LangSmith, LangFuse, W&B, FastAPI, PyTorch, Docker, Git, CI/CD, GitHub Actions.

Cloud & Infrastructure: SLURM, AWS, Azure, GCP (limited), Docker-Compose, Kubernetes, Terraform

Core Competencies: Agentic & Multi-Agent Systems (LangGraph, MCP & tool-using agents, orchestration, fallback/recovery), Production AI Systems (observability, evaluation pipelines, model routing, cost/latency optimization, monitoring, rollback), RAG, LLM Integration, API Design, Scalable Backend Systems

PROJECTS

Assessing LLM Agents in Real-World Automation Environments, Jan 2025 – Sep 2025
M.Sc. Thesis – Universitaet Potsdam
- Built CoLabGame, a Python evaluation framework benchmarking single- and multi-agent LLMs on real-world computer-use tasks across OSWorld desktop environments.
- Designed structured role-based (agent-skills) collaboration topologies/ architecture and orchestration protocols for LLM agents operating over GUI, terminal, and multimodal interfaces.
- Ran large-scale experiments across frontier models (mainly GPT-5, GPT-5-mini, Gemini-2.5, Claude-4-Sonnet), analyzing execution success, coordination efficiency, failure modes, and operational cost trade-offs.
- Demonstrated that centralized hierarchical agent architectures outperform decentralized and single-agent systems on complex workflows in robustness, efficiency, and cost-effectiveness.

Emergent Abilities in cLLMs, Oct 2023 – Jun 2024
Independent Research Module – Universitaet Potsdam
- Built an end-to-end experimental study evaluating LLM (as agents) behavior in self-play conversational tasks, including experiment design, model configuration steps, and structured analysis of outcome across realistic inference settings.
- Found that increased model scale does not consistently translate to improved conversational performance, revealing reliability & cost-performance trade-offs relevant to production model selection.

Interaction is All You Need - Embodied Agent System, Apr 2023 – Sep 2023
Project Module – Universitaet Potsdam
- Built and benchmarked multimodal transformer-based embodied agents for household task execution, integrating ViT vision encoders with pretrained sequence-to-sequence language models.
- PEFT finetuned text-only LLaMA2 LLM, proving that scaling-up can partially compensate for missing visual context in action planning.

FewJIE-BERT, Research Project - DFKI, Berlin, Jan 2023 – Feb 2024
- Built a BERT-based joint model for entity recognition and relation extraction, examining unified architectures to improve consistency in document-level NLP.
- Investigated few-shot learning for structured prediction using metric learning and clustering-based inference to adapt to new entity and relation types with limited data.
- Stack: PyTorch, HuggingFace Transformers, BERT, episodic training, custom losses (contrastive, focal, Dice), FewNERD & DocRED datasets, Weights & Biases for experiment tracking, and Gradio demos.

PUBLICATIONS

How Many Parameters Does it Take to Change a Light Bulb? Evaluating Performance in Self-Play of Conversational Games as a Function of Model Characteristics, ArXiv Preprint — 2024

Interaction is all You Need? A Study of Robots Ability to Understand and Execute, Arxiv Preprint — 2023

HMIST: Hierarchical Multilingual Isometric Speech Translation using Multi-Task Learning Framework and it's influence on Automatic Dubbing., Long Description Paper at PACLIC 36. — 2022

DeepCon: An End-to-End Multilingual Toolkit for Automatic Minuting of Multi-Party Dialogues, Demo paper at SIGDIAL 2022. — 2022

An End-to-End Multilingual System for Automatic Minuting of Multi-Party Dialogues, Long Description Paper at PACLIC 36. — 2022

Team ABC System Run for AutoMin Shared Task, — 2021
In Proceedings of the AutoMin workshop at ISCA Interspeech 2021.